BST Physics Curriculum Vol 12 Ch 2 — Chemical Bonding from Substrate v0.4 (Calibration #23 substance-floor refill)

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Vol 12 Chapter 2 — Chemical Bonding from Substrate

Chapter motivation

Chemical bonds organize the periodic table into molecules. Standard treatment (Atkins, McQuarrie, Pauling): orbital overlap + Pauli exclusion + electronegativity differences → four bond classes (covalent + ionic + hydrogen + van der Waals/dispersion). BST substrate-cartography reading: each bond class is a specific substrate K-type tensor product structure. The atomic orbital sequence (2l+1) = 1, 3, 5, 7 reads the BST primary integer ladder (1, N_c, n_C, g). Bonding chemistry is substrate sub-structure coupling cataloged at the molecular scale.

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: Chemical bonds = substrate K-type tensor product structures. Covalent + ionic + hydrogen + van der Waals = four bond classes from four substrate coupling modes. Atomic orbital ladder (2l+1) = 1, 3, 5, 7 reads BST primary integers (1, N_c, n_C, g).

Level 2 (graduate-chemist accessible): Standard chemical bonding theory: covalent bonds via shared electron pairs (Lewis 1916, Pauling 1939 The Nature of the Chemical Bond), ionic bonds via electronegativity differences > 1.7 (Pauling scale), hydrogen bonds via partial-charge attraction (Latimer-Rodebush 1920), van der Waals/dispersion via instantaneous dipole correlation (London 1930). Quantum-mechanical foundation: molecular orbital (MO) theory (Hund-Mulliken 1928) and valence bond (VB) theory (Heitler-London 1927); these are complementary descriptions of the same underlying multi-electron wavefunction. BST substrate-cartography reading: each bond class corresponds to a specific substrate K-type tensor product structure on D_IV^5. Atomic orbital quantum numbers (n, l, m_l, m_s) classify Wallach K-type representations; angular momentum sequence (2l+1) for l = 0, 1, 2, 3 gives the integer sequence (1, 3, 5, 7) which reads (1, N_c, n_C, g) — the BST primary integer ladder. Covalent bonding via orbital overlap = substrate K-type tensor product (K_A ⊗ K_B → K_AB symmetric); ionic bonding via electron transfer = substrate K-type asymmetric coupling (Z_eff differential); hydrogen bonding via proton siblings = substrate proton-mass m_p/m_e = $6\pi^5$ cross-link to Vol 13 Ch 4; van der Waals via dispersion = substrate vacuum sub-structure coupling (Casimir cross-link Vol 4 Ch 4 Λ via T2418 unification). Bond energies in standard chemistry texts (e.g., C-H bond 413 kJ/mol, C=C double bond 614 kJ/mol, C≡C triple bond 839 kJ/mol) emerge from substrate K-type Casimir spectrum eigenvalues at molecular configurations.

Level 3 (5th-grader accessible): Atoms stick together to make molecules using four kinds of "glue": (1) covalent — they share electrons (most chemistry); (2) ionic — one atom steals an electron from another (salt); (3) hydrogen — small partial-charge attraction (water, DNA); (4) van der Waals — tiny temporary attractions (geckos walking on ceilings). BST framework says all four glue types come from the substrate D_IV^5 combining its building blocks (called "K-types") in different ways. The pattern (1, 3, 5, 7) you see in atomic orbitals is the same BST integer pattern (1, N_c, n_C, g). Chemistry is substrate-cartography at the molecular scale.

Section 2.1 — Covalent Bonding as K-Type Tensor Product

Covalent bonds form when two atomic orbitals overlap and electrons pair to occupy the resulting molecular orbital. BST framing: K-type representations K_A on atom A and K_B on atom B combine via tensor product K_A ⊗ K_B; the symmetric (bonding) and antisymmetric (antibonding) projections correspond to bonding and antibonding molecular orbitals. C-H, C-C, C=C, C≡C bond strengths and lengths fall in standard ranges: single bond ~150 pm + 350-400 kJ/mol; double bond ~134 pm + 600-700 kJ/mol; triple bond ~120 pm + 800-900 kJ/mol. Per Toy 1789 + Vol 12 Ch 3 cross-link, bond-energy ratios single:double:triple $\approx 1 : 1.7 : 2.3$ fall close to BST primary ratios.

Section 2.2 — Ionic Bonding via Electronegativity

Ionic bonds form when electronegativity difference $\Delta\chi > 1.7$ (Pauling scale); electron transfer creates cation + anion pair with electrostatic attraction. BST framing: asymmetric K-type coupling where one substrate sub-structure (anion) absorbs charge and the other (cation) loses it. Lattice energies for ionic crystals (NaCl ~787 kJ/mol, MgO ~3795 kJ/mol) emerge from substrate K-type coupling strengths weighted by ionic charges. Per Vol 12 Ch 1 periodic table reading: electronegativity gradient across periods reflects substrate Z_eff scaling derivable from BST integer structure.

Section 2.3 — Hydrogen Bonding and Proton Siblings

Hydrogen bonds (5-30 kJ/mol; longer than covalent at ~250-350 pm) are partial-charge attractions involving H atoms covalently bound to electronegative atoms (O, N, F). Critical for water structure, DNA base pairing, protein secondary structure. BST framing: hydrogen bonding is a substrate proton-electron mass-ratio phenomenon — proton mass m_p/m_e = $6\pi^5$ (BST T185 D-tier 0.002% precision; Vol 1 Ch 11 reference compilation). Proton-electron mass ratio constrains substrate molecular vibrational modes. Vol 13 Ch 4 DNA Proton Siblings extends this to biological structures.

Section 2.4 — Van der Waals / Dispersion Forces

Van der Waals forces (0.1-10 kJ/mol; ~300-500 pm range) include London dispersion (instantaneous dipole correlation), dipole-dipole, and dipole-induced-dipole. London dispersion energies depend on polarizability and follow ~$1/r^6$ distance dependence. BST framing: dispersion forces are substrate vacuum sub-structure couplings — the same substrate vacuum that produces cosmological Λ at Zone-4 (Vol 4 Ch 4) and Casimir force at Zone-2 (T2418 unification, Wednesday) produces intermolecular dispersion at Zone-3 (molecular scale). Universal mechanism across 122-order energy hierarchy.

Section 2.5 — Connection

- Vol 12 Ch 1 Periodic Table from Substrate (atomic orbital ladder reading)
- Vol 3 Ch 7 Atomic Orbitals (2l+1 = 1, N_c, n_C, g)
- Vol 5 QM K-type representation theory
- Vol 13 Ch 4 DNA Proton Siblings (hydrogen bonding biological role)
- Vol 4 Ch 4 + T2418 (substrate vacuum unification dispersion ↔ Casimir ↔ Λ)
- Vol 12 Ch 3 Molecular Orbital Theory (next chapter; MO/VB formal treatment)

Honest scope: Substrate-mechanism for each bond class is identified at framework level; specific quantitative bond-energy derivations from substrate K-type Casimir spectrum remain multi-year program parallel to SP-31 substrate-Hamiltonian closure. Bond ratios single:double:triple match BST integer-product structure at I-tier.

— Lyra, Vol 12 Ch 2 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT